Lecture 7

Cartan subalgebras and root decomposition

Objectives. *We use simultaneous eigenspaces to study the structure of a semisimple Lie algebra. We take a maximal subalgebra of elements x with $\operatorname{ad} x$ diagonalisable. Its adjoint eigenspaces for nonzero weights are the root spaces. The Killing form pairs opposite root spaces.*

By the end of this lecture you will be able to:

- Prove that toral subalgebras are abelian and that Cartan subalgebras exist.
- Derive the root decomposition and the rule $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$.
- Prove the Killing-form orthogonality relations.
- Compute the roots, root matrices and brackets of $\mathfrak{sl}_3(\mathbb{C})$.

Throughout, $\mathfrak{g}$ is a nonzero semisimple complex Lie algebra with Killing form κ. By Lecture 4, κ is symmetric, nondegenerate and invariant:

$$\kappa([x,y],z) = \kappa(x,[y,z]) \quad \text{for all } x,y,z \in \mathfrak{g}.$$

The centre $Z(\mathfrak{g})$ is an abelian ideal. Abelian ideals are solvable, so $Z(\mathfrak{g})$ lies in the radical, which is zero. Hence

$$Z(\mathfrak{g}) = 0.$$

7.1 Toral subalgebras and imports

An element $x \in \mathfrak{g}$ is *semisimple* if $\operatorname{ad} x$ is diagonalisable. The element x is *nilpotent* if $\operatorname{ad} x$ is nilpotent. Lecture 4 stated the internal Jordan decomposition [C] (Humphreys §5.4): each $x \in \mathfrak{g}$ has a unique expression

$$x = x_s + x_n, \qquad x_s, x_n \in \mathfrak{g}, \qquad [x_s, x_n] = 0,$$

with x_s semisimple and x_n nilpotent. If x is semisimple, then $x = x + 0$ is such an expression. Uniqueness gives $x_s = x$ and $x_n = 0$.

Definition 7.1 (Toral and Cartan subalgebras)**.** A subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$ is *toral* if every element of $\mathfrak{t}$ is semisimple. A *Cartan subalgebra* of $\mathfrak{g}$ is a toral subalgebra not properly contained in another toral subalgebra.

The zero subspace is toral. Every toral subalgebra has dimension at most $\dim \mathfrak{g}$, so it lies in a maximal one. Hence Cartan subalgebras exist.

The next proof uses a fact from linear algebra. Let T be a diagonalisable operator with distinct eigenvalues $c_1, \dots, c_r$, and let U be a T-invariant subspace. Then

$$(T - c_1) \cdots (T - c_r) = 0 \quad \Longrightarrow \quad (T|_U - c_1) \cdots (T|_U - c_r) = 0.$$

So the minimal polynomial of $T|_U$ divides a product of distinct linear factors. Hence $T|_U$ is diagonalisable.

Lemma 7.2 (Toral subalgebras are abelian · *[P]*)**.** *Every toral subalgebra $\mathfrak{t}$ of $\mathfrak{g}$ is abelian.*

Proof. Fix $x \in \mathfrak{t}$. Let $\operatorname{ad}_{\mathfrak{t}} x$ be the restriction of $\operatorname{ad} x$ to the invariant subspace $\mathfrak{t}$. By the fact above, this restriction is diagonalisable. A diagonalisable operator whose only eigenvalue is 0 is zero. So it suffices to show that $\operatorname{ad}_{\mathfrak{t}} x$ has no nonzero eigenvalue.

Suppose that $y \in \mathfrak{t}$ is nonzero and $[x, y] = ay$ with $a \neq 0$. Apply $\operatorname{ad}_{\mathfrak{t}} y$ twice:

$$(\operatorname{ad}_{\mathfrak{t}} y)(x) = [y, x] = -ay,$$
$$(\operatorname{ad}_{\mathfrak{t}} y)(-ay) = -a[y, y] = 0.$$

Thus the vector $[y, x]$ lies in the kernel of $\operatorname{ad}_{\mathfrak{t}} y$.

The operator $\operatorname{ad}_{\mathfrak{t}} y$ is diagonalisable as well. Decompose x into eigenvectors of $\operatorname{ad}_{\mathfrak{t}} y$:

$$x = \sum_{\lambda} x_\lambda, \qquad [y, x_\lambda] = \lambda x_\lambda.$$

Applying $\operatorname{ad}_{\mathfrak{t}} y$ gives

$$[y, x] = \sum_{\lambda \neq 0} \lambda x_\lambda.$$

The right side lies in the sum of the nonzero eigenspaces. That sum meets the kernel only in 0. Hence

$$-ay = [y, x] = 0.$$

Since $a \neq 0$ and $y \neq 0$, this is a contradiction. ∎

We import two theorems here, and a third in Section 7.5. The second is used only in Corollary 7.9.

Theorem 7.3 (Self-centraliser theorem · *[C]*)**.** *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and let $\mathfrak{h}$ be a Cartan subalgebra. Then*

$$C_{\mathfrak{g}}(\mathfrak{h}) = \{x \in \mathfrak{g} : [h, x] = 0 \text{ for all } h \in \mathfrak{h}\} = \mathfrak{h}.$$

Source: Humphreys §8.2, Proposition, for a semisimple algebra over an algebraically closed field of characteristic zero. The inclusion $\mathfrak{h} \subseteq C_{\mathfrak{g}}(\mathfrak{h})$ is Lemma 7.2. The reverse inclusion is the imported part.

Hence every Cartan subalgebra is nonzero, because the zero subalgebra has centraliser $C_{\mathfrak{g}}(0) = \mathfrak{g} \neq 0$.

Theorem 7.4 (Jordan decomposition in representations · *[C]*)**.** *Let $\mathfrak{g}$ be semisimple, let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation, and let $x = x_s + x_n$ be the internal Jordan decomposition of $x \in \mathfrak{g}$. Then $\rho(x_s)$ is diagonalisable, $\rho(x_n)$ is nilpotent, and $\rho(x) = \rho(x_s) + \rho(x_n)$ is the Jordan decomposition of the operator $\rho(x)$.*

Source: Humphreys §6.4, Corollary. Its proof uses Weyl's theorem from Lecture 5.

7.2 The root decomposition

Lemma 7.5 (Simultaneous eigenspaces · *[P]*). *Let $\mathfrak{a}$ be an abelian Lie algebra, and let ρ be a representation of $\mathfrak{a}$ on a finite-dimensional space V such that every $\rho(a)$ is diagonalisable. For $\mu \in \mathfrak{a}^*$ put*

$$V_\mu = \{v \in V : \rho(a)v = \mu(a)v \text{ for all } a \in \mathfrak{a}\}.$$

Then $V = \bigoplus_{\mu \in \mathfrak{a}^*} V_\mu$, and only finitely many summands are nonzero.

Proof. The operators commute, because $\mathfrak{a}$ is abelian:

$$\rho(a)\rho(b) - \rho(b)\rho(a) = \rho([a,b]) = 0.$$

Choose a basis $T_1, \dots, T_m$ of the finite-dimensional space $\rho(\mathfrak{a})$. These commuting diagonalisable operators have a common eigenbasis $(v_1, \dots, v_d)$. It is an eigenbasis for every $\rho(a)$, since each $\rho(a)$ is a linear combination of the T_i. Write $\rho(a)v_k = \mu_k(a)v_k$. For $a, b \in \mathfrak{a}$ and $s, t \in \mathbb{C}$,

$$\begin{aligned} \mu_k(sa+tb)\,v_k &= \rho(sa+tb)v_k \\ &= s\rho(a)v_k + t\rho(b)v_k = \big(s\mu_k(a) + t\mu_k(b)\big)v_k. \end{aligned}$$

Thus each μ_k is linear, $v_k \in V_{\mu_k}$, and the spaces V_μ span V.

For directness, let $v = \sum_k c_k v_k \in V_\mu$. For every $a \in \mathfrak{a}$,

$$0 = \big(\rho(a) - \mu(a)\big)v = \sum_{k=1}^{d} c_k \big(\mu_k(a) - \mu(a)\big)v_k.$$

So $c_k = 0$ whenever $\mu_k \neq \mu$. Thus V_μ is spanned by the v_k with $\mu_k = \mu$. The sum is direct, and only the forms μ_k occur. ∎

Fix a Cartan subalgebra $\mathfrak{h}$. It is abelian by Lemma 7.2, and every $\operatorname{ad} h$ with $h \in \mathfrak{h}$ is diagonalisable. Apply Lemma 7.5 to the representation $\mathfrak{h} \to \mathfrak{gl}(\mathfrak{g})$, $h \mapsto \operatorname{ad} h$.

Definition 7.6 (Roots). For $\alpha \in \mathfrak{h}^*$, put

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} : [h,x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}.$$

A *root* is a nonzero $\alpha \in \mathfrak{h}^*$ with $\mathfrak{g}_\alpha \neq 0$. For a root α, the space $\mathfrak{g}_\alpha$ is its *root space*. We write R for the set of roots.

Theorem 7.7 (Root decomposition · *[P]*). *The set R is finite, and*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

Proof. Lemma 7.5 gives

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha,$$

with R finite. By definition, $\mathfrak{g}_0$ is the centraliser of $\mathfrak{h}$. Theorem 7.3 identifies it:

$$\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$ ∎

Proposition 7.8 (Bracket rule · *[P]*). *For all $\alpha, \beta \in \mathfrak{h}^*$,*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

Proof. Let $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$. The Jacobi identity in derivation form (Lecture 1) gives

$$\begin{aligned}[h,[x,y]] &= [[h,x],y] + [x,[h,y]] \\ &= \alpha(h)[x,y] + \beta(h)[x,y] = (\alpha+\beta)(h)\,[x,y].\end{aligned}$$

■

Three consequences follow.

(i) If $\alpha + \beta \notin R \cup \{0\}$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = 0$.

(ii) If $\alpha \in R$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{g}_0 = \mathfrak{h}$.

(iii) If $\alpha \in R$ and $x \in \mathfrak{g}_\alpha$, then $\operatorname{ad} x$ is nilpotent.

For (iii), apply the bracket rule k times:

$$(\operatorname{ad} x)^k \mathfrak{g}_\beta \subseteq \mathfrak{g}_{\beta + k\alpha} \quad \text{for all } k \geq 0.$$

Since $\alpha \neq 0$, the forms $\beta + k\alpha$ with $k \geq 0$ are distinct. Only finitely many lie in the finite set $R \cup \{0\}$. So for each $\beta \in R \cup \{0\}$ there is N_β with $(\operatorname{ad} x)^k \mathfrak{g}_\beta = 0$ for $k \geq N_\beta$. Let N be the largest N_β. By the root decomposition,

$$(\operatorname{ad} x)^N \mathfrak{g} = \sum_{\beta \in R \cup \{0\}} (\operatorname{ad} x)^N \mathfrak{g}_\beta = 0.$$

Corollary 7.9 (Weights of a module · *[P] given Theorem 7.4*)**.** *Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation. Define V_μ as in Lemma 7.5 for ρ restricted to $\mathfrak{h}$. Then*

$$V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu, \qquad \rho(\mathfrak{g}_\alpha) V_\mu \subseteq V_{\mu+\alpha} \quad \text{for all } \alpha, \mu \in \mathfrak{h}^*.$$

Proof. For $h \in \mathfrak{h}$ we have $h_s = h$, so $\rho(h)$ is diagonalisable by Theorem 7.4. Lemma 7.5 gives the decomposition.

For $x \in \mathfrak{g}_\alpha$ and $v \in V_\mu$, use $\rho([h,x]) = \alpha(h)\rho(x)$:

$$\begin{aligned}\rho(h)\rho(x)v &= \rho(x)\rho(h)v + \rho([h,x])v \\ &= \mu(h)\rho(x)v + \alpha(h)\rho(x)v = \big(\mu+\alpha\big)(h)\,\rho(x)v.\end{aligned}$$

■

The forms μ with $V_\mu \neq 0$ are the *weights* of V. For the adjoint representation they are 0 and the roots.

7.3 Killing-form orthogonality

Proposition 7.10 (Orthogonality · *[P]*)**.**

(a) If $\alpha, \beta \in \mathfrak{h}^*$ and $\alpha + \beta \neq 0$, then $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$.

(b) The restriction of κ to $\mathfrak{h}$ is nondegenerate.

(c) If $\alpha \in R$, then $-\alpha \in R$, and κ restricts to a nondegenerate pairing $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$.

Proof. (a) Let $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$. By antisymmetry and invariance,

$$\begin{aligned}\alpha(h)\,\kappa(x,y) &= \kappa([h,x],y) = -\kappa([x,h],y) \\ &= -\kappa(x,[h,y]) = -\beta(h)\,\kappa(x,y).\end{aligned}$$

Thus $(\alpha+\beta)(h)\,\kappa(x,y) = 0$ for every $h \in \mathfrak{h}$. Since $\alpha+\beta \neq 0$, some h has $(\alpha+\beta)(h) \neq 0$. Hence $\kappa(x,y) = 0$.

(b) Let $z \in \mathfrak{h}$ satisfy $\kappa(z, \mathfrak{h}) = 0$. Since $z \in \mathfrak{g}_0$, part (a) gives $\kappa(z, \mathfrak{g}_\beta) = 0$ for every root β. Take $y \in \mathfrak{g}$ and decompose it by Theorem 7.7:

$$y = y_0 + \sum_{\beta \in R} y_\beta, \qquad y_0 \in \mathfrak{h}, \quad y_\beta \in \mathfrak{g}_\beta.$$

Then

$$\kappa(z, y) = \kappa(z, y_0) + \sum_{\beta \in R} \kappa(z, y_\beta) = 0.$$

Nondegeneracy of κ on $\mathfrak{g}$ gives $z = 0$.

(c) Let $x \in \mathfrak{g}_\alpha$ be nonzero. Nondegeneracy of κ gives $y \in \mathfrak{g}$ with $\kappa(x, y) \neq 0$. Decompose y as in (b). By (a), $\kappa(x, y_0) = 0$ because $\alpha \neq 0$, and $\kappa(x, y_\beta) = 0$ unless $\beta = -\alpha$. If $-\alpha \notin R$, every term vanishes, which is impossible. So $-\alpha \in R$, and

$$0 \neq \kappa(x, y) = \kappa(x, y_0) + \sum_{\beta \in R} \kappa(x, y_\beta) = \kappa(x, y_{-\alpha}).$$

Hence no nonzero element of $\mathfrak{g}_\alpha$ is orthogonal to $\mathfrak{g}_{-\alpha}$. Exchanging α and $-\alpha$ proves nondegeneracy on the other side. ∎

A nondegenerate pairing identifies each space with the dual of the other. Hence $\dim \mathfrak{g}_\alpha = \dim \mathfrak{g}_{-\alpha}$.

Since κ is nondegenerate on $\mathfrak{h}$, the linear map

$$\mathfrak{h} \longrightarrow \mathfrak{h}^*, \qquad t \longmapsto \kappa(t, \cdot)$$

has zero kernel. Both spaces have dimension $\dim \mathfrak{h}$, so the map is bijective. For $\lambda \in \mathfrak{h}^*$, let $t_\lambda \in \mathfrak{h}$ be the unique element with

$$\kappa(t_\lambda, h) = \lambda(h) \quad \text{for all } h \in \mathfrak{h}.$$

Proposition 7.11 (Roots and the Killing form · *[P]*)**.**

(a) The roots span $\mathfrak{h}^*$.

(b) If $\alpha \in R$, $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$, then $[x, y] = \kappa(x, y)\, t_\alpha$. Hence $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C} t_\alpha$.

Proof. (a) Suppose the roots span a proper subspace of $\mathfrak{h}^*$. By duality, some nonzero $h \in \mathfrak{h}$ satisfies $\alpha(h) = 0$ for all $\alpha \in R$. Then

$$[h, x] = \alpha(h) x = 0 \qquad \text{for all } x \in \mathfrak{g}_\alpha,\ \alpha \in R.$$

Also $[h, \mathfrak{h}] = 0$, since $\mathfrak{h}$ is abelian. By Theorem 7.7, these subspaces span $\mathfrak{g}$. So $h \in Z(\mathfrak{g}) = 0$, a contradiction.

(b) By the bracket rule, $[x, y] \in \mathfrak{g}_0 = \mathfrak{h}$. For $h \in \mathfrak{h}$, invariance and the definition of t_α give

$$\begin{aligned} \kappa(h, [x, y]) &= \kappa([h, x], y) = \alpha(h)\, \kappa(x, y) \\ &= \kappa(t_\alpha, h)\, \kappa(x, y) = \kappa\big(h, \kappa(x, y)\, t_\alpha\big). \end{aligned}$$

Both $[x, y]$ and $\kappa(x, y) t_\alpha$ lie in $\mathfrak{h}$. Their difference is orthogonal to $\mathfrak{h}$. Since κ is nondegenerate on $\mathfrak{h}$ by Proposition 7.10(b), the difference is zero.

By Proposition 7.10(c), some $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ have $\kappa(x, y) \neq 0$. Since $\alpha \neq 0$, also $t_\alpha \neq 0$. These brackets therefore span $\mathbb{C} t_\alpha$. ∎

Lecture 8 starts from Proposition 7.11(b). It attaches an $\mathfrak{sl}_2$-subalgebra to each root and proves $\dim \mathfrak{g}_\alpha = 1$.

7.4 Worked example: $\mathfrak{sl}_3(\mathbb{C})$

Let $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, and let $\mathfrak{h}$ be the traceless diagonal matrices, with basis

$$h_1 = E_{11} - E_{22}, \qquad h_2 = E_{22} - E_{33}.$$

Take $H = \operatorname{diag}(a_1, a_2, a_3)$ in $\mathfrak{h}$ and $X \in \mathfrak{g}$. The entries of the two products are

$$(HX)_{ij} = a_i X_{ij}, \qquad (XH)_{ij} = X_{ij} a_j.$$

Subtracting gives

$$[H, X]_{ij} = (a_i - a_j) X_{ij}, \qquad [H, E_{ij}] = (a_i - a_j) E_{ij}.$$

Also $[H, h_1] = [H, h_2] = 0$. The eight matrices h_1, h_2 and E_{ij} with $i \neq j$ form an eigenbasis of $\operatorname{ad} H$. Hence $\mathfrak{h}$ is toral.

Next, $\mathfrak{h}$ is a Cartan subalgebra. Put $H_0 = \operatorname{diag}(1, 0, -1)$. Its entry differences are

$$a_1 - a_2 = 1, \qquad a_1 - a_3 = 2, \qquad a_2 - a_3 = 1.$$

None is zero. If $X \in C_{\mathfrak{g}}(\mathfrak{h})$, then $[H_0, X] = 0$, so $X_{ij} = 0$ for $i \neq j$. Thus X is diagonal and traceless, and

$$C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

If $\mathfrak{t} \supseteq \mathfrak{h}$ is toral, Lemma 7.2 gives $\mathfrak{t} \subseteq C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. So $\mathfrak{t} = \mathfrak{h}$.

The proofs in §7.2 and §7.3 use Theorem 7.3 only through $\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. The required equality has been checked directly here.

Define $\varepsilon_i \in \mathfrak{h}^*$ by $\varepsilon_i(H) = a_i$. On $\mathfrak{h}$, $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$. The eigenvector E_{ij} has weight $\varepsilon_i - \varepsilon_j$. Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$, so that $\varepsilon_1 - \varepsilon_3 = \alpha_1 + \alpha_2$. Two representative values are

$$\alpha_1(h_1) = \varepsilon_1(h_1) - \varepsilon_2(h_1) = 1 - (-1) = 2,$$
$$\alpha_2(h_1) = \varepsilon_2(h_1) - \varepsilon_3(h_1) = -1 - 0 = -1.$$

The other values are found in the same way. The table lists the six weights, their root matrices and their values.

root	α_1	α_2	$\alpha_1 + \alpha_2$	$-\alpha_1$	$-\alpha_2$	$-\alpha_1 - \alpha_2$
root matrix	E_{12}	E_{23}	E_{13}	E_{21}	E_{32}	E_{31}
value at h_1	2	-1	1	-2	1	-1
value at h_2	-1	2	1	1	-2	-1

The table gives distinct nonzero value pairs. So the eigenbasis gives

$$R = \{\varepsilon_i - \varepsilon_j : i \neq j\}, \qquad \mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{C} E_{ij}, \qquad \dim \mathfrak{g} = 2 + 6.$$

Here one-dimensionality is read off from a basis. Lecture 8 proves it for every semisimple $\mathfrak{g}$.

The matrix-unit rule $E_{ij} E_{kl} = \delta_{jk} E_{il}$ gives four nonzero brackets:

$$[E_{12}, E_{23}] = E_{12}E_{23} - E_{23}E_{12} = E_{13} - 0 = E_{13},$$
$$[E_{13}, E_{32}] = E_{13}E_{32} - E_{32}E_{13} = E_{12} - 0 = E_{12},$$
$$[E_{23}, E_{31}] = E_{23}E_{31} - E_{31}E_{23} = E_{21} - 0 = E_{21},$$
$$[E_{12}, E_{21}] = E_{12}E_{21} - E_{21}E_{12} = E_{11} - E_{22} = h_1.$$

Proposition 7.8 predicts that each result has the sum of the two input weights. In order, these sums are

$$\alpha_1 + \alpha_2, \qquad (\alpha_1 + \alpha_2) - \alpha_2 = \alpha_1,$$
$$\alpha_2 - (\alpha_1 + \alpha_2) = -\alpha_1, \qquad \alpha_1 - \alpha_1 = 0.$$

They are the weights of E_{13}, E_{12}, E_{21} and h_1.

By contrast,

$$[E_{12}, E_{13}] = E_{12}E_{13} - E_{13}E_{12} = 0 - 0 = 0.$$

The bracket rule requires this, since the bracket lies in $\mathfrak{g}_{2\alpha_1+\alpha_2}$. The sum of the two weights takes the values

$$(2\alpha_1 + \alpha_2)(h_1) = 2 \cdot 2 - 1 = 3, \qquad (2\alpha_1 + \alpha_2)(h_2) = 2 \cdot (-1) + 2 = 0.$$

These values are not $(0, 0)$, and no column of the table has the values $(3, 0)$. So $2\alpha_1 + \alpha_2 \notin R \cup \{0\}$, and $\mathfrak{g}_{2\alpha_1+\alpha_2} = 0$.

Now compute κ on $\mathfrak{h}$. Take $H = \operatorname{diag}(a_i)$ and $H' = \operatorname{diag}(b_i)$ in $\mathfrak{h}$. The operator $\operatorname{ad} H \operatorname{ad} H'$ is zero on $\mathfrak{h}$ and multiplies E_{ij} by $(a_i - a_j)(b_i - b_j)$. Its trace in the eigenbasis is

$$\kappa(H, H') = \sum_{i \neq j} (a_i - a_j)(b_i - b_j) = \sum_{i,j=1}^{3} (a_i - a_j)(b_i - b_j),$$

since the terms with $i = j$ are zero. Expand the product and sum over i and j:

$$\begin{aligned}\sum_{i,j=1}^{3} (a_i - a_j)(b_i - b_j) &= \sum_{i,j=1}^{3} (a_i b_i + a_j b_j) - \sum_{i,j=1}^{3} (a_i b_j + a_j b_i) \\ &= 6 \sum_{i=1}^{3} a_i b_i - 2\Big(\sum_{i=1}^{3} a_i\Big)\Big(\sum_{j=1}^{3} b_j\Big) \\ &= 6 \sum_{i=1}^{3} a_i b_i.\end{aligned}$$

The last step uses $\sum_i a_i = \sum_i b_i = 0$.

Let $\alpha = \varepsilon_i - \varepsilon_j$ with $i \neq j$. The matrix $\frac{1}{6}(E_{ii} - E_{jj})$ lies in $\mathfrak{h}$, and

$$\kappa\Big(\frac{1}{6}(E_{ii} - E_{jj}), H\Big) = 6\Big(\frac{1}{6}a_i - \frac{1}{6}a_j\Big) = a_i - a_j = \alpha(H).$$

By uniqueness of t_α,

$$t_\alpha = \frac{1}{6}(E_{ii} - E_{jj}), \qquad t_{\alpha_1} = \frac{1}{6}h_1.$$

Proposition 7.11(b) now gives a Killing value without a trace:

$$h_1 = [E_{12}, E_{21}] = \kappa(E_{12}, E_{21})\, t_{\alpha_1} = \frac{1}{6}\kappa(E_{12}, E_{21})\, h_1.$$

Since $h_1 \neq 0$, we get $\kappa(E_{12}, E_{21}) = 6$. Proposition 7.10(a) gives other values with no calculation:

$$\kappa(E_{12}, E_{12}) = 0 \quad \text{since } 2\alpha_1 \neq 0,$$
$$\kappa(E_{12}, E_{23}) = 0 \quad \text{since } \alpha_1 + \alpha_2 \neq 0.$$

7.5 Changing the Cartan subalgebra

The root decomposition depends on the choice of $\mathfrak{h}$. An automorphism φ of $\mathfrak{g}$ carries the decomposition for $\mathfrak{h}$ to the decomposition for $\varphi(\mathfrak{h})$. For $x, y \in \mathfrak{g}$,

$$(\operatorname{ad}\varphi(x))(\varphi(y)) = [\varphi(x), \varphi(y)] = \varphi([x, y]), \quad \text{so} \quad \operatorname{ad}\varphi(x) = \varphi \circ \operatorname{ad} x \circ \varphi^{-1}.$$

Hence $\operatorname{ad}\varphi(x)$ is diagonalisable exactly when $\operatorname{ad} x$ is. The product $\operatorname{ad}\varphi(x)\operatorname{ad}\varphi(y)$ is conjugate to $\operatorname{ad} x \operatorname{ad} y$, so taking traces gives $\kappa(\varphi x, \varphi y) = \kappa(x, y)$.

Thus φ and φ^{-1} map toral subalgebras to toral subalgebras. If $\mathfrak{t}' \supseteq \varphi(\mathfrak{h})$ is toral, then $\varphi^{-1}(\mathfrak{t}') \supseteq \mathfrak{h}$ is toral. By maximality, $\varphi^{-1}(\mathfrak{t}') = \mathfrak{h}$, so $\mathfrak{t}' = \varphi(\mathfrak{h})$. Hence $\mathfrak{h}' = \varphi(\mathfrak{h})$ is a Cartan subalgebra.

For $x \in \mathfrak{g}_\alpha$ and $h \in \mathfrak{h}$,

$$[\varphi(h), \varphi(x)] = \varphi([h, x]) = \alpha(h)\,\varphi(x).$$

Put $\alpha' = \alpha \circ (\varphi|_{\mathfrak{h}})^{-1} \in (\mathfrak{h}')^*$. Hence $\varphi(\mathfrak{g}_\alpha)$ lies in the root space of $\mathfrak{h}'$ for α'. The same argument for φ^{-1} gives equality. Roots, brackets and Killing values are carried over unchanged.

> **Theorem 7.12** (Conjugacy of Cartan subalgebras · *[C]*). *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and let $\mathfrak{h}$ and $\mathfrak{h}'$ be Cartan subalgebras. Then some automorphism φ of $\mathfrak{g}$ satisfies $\varphi(\mathfrak{h}) = \mathfrak{h}'$.*

Source: Humphreys §16.4, Corollary (conjugacy under the group $E(L)$). Humphreys defines Cartan subalgebras as nilpotent self-normalising subalgebras. By Humphreys §15.3, Corollary, these are exactly the maximal toral subalgebras when $\mathfrak{g}$ is semisimple.

The theorem and the transport argument show that the root decompositions of any two Cartan subalgebras are related by an automorphism of $\mathfrak{g}$. Their common dimension is the *rank* of $\mathfrak{g}$. Thus the classification in Lecture 10 does not depend on the choice.

Self-checks

(1) Use the basis (h, e, f) of $\mathfrak{sl}_2(\mathbb{C})$ from Lecture 1. Is the line $\mathbb{C}e$ toral?

(2) In $\mathfrak{sl}_3(\mathbb{C})$, which root space pairs nondegenerately with $\mathfrak{g}_{\alpha_1+\alpha_2}$? What is $[E_{13}, E_{31}]$?

(3) Can $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta]$ be nonzero when $\alpha + \beta \notin R \cup \{0\}$?

Answers. (1) No. Since $(\operatorname{ad} e)^2 f = [e, h] = -2e \neq 0$ and $(\operatorname{ad} e)^3 = 0$, the operator $\operatorname{ad} e$ is nonzero and nilpotent, hence not diagonalisable.
(2) The space $\mathfrak{g}_{-\alpha_1-\alpha_2} = \mathbb{C}E_{31}$. The bracket is

$$[E_{13}, E_{31}] = E_{11} - E_{33} = h_1 + h_2 = 6\,t_{\alpha_1+\alpha_2},$$

so $\kappa(E_{13}, E_{31}) = 6$.
(3) No. Such a bracket lies in $\mathfrak{g}_{\alpha+\beta} = 0$. For example, $[E_{12}, E_{13}] = 0$.

Summary

A Cartan subalgebra is a maximal toral subalgebra. It is abelian, and by the cited self-centraliser theorem it equals its centraliser. Its adjoint eigenspaces give

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad [\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

The Killing form pairs $\mathfrak{g}_\alpha$ only with $\mathfrak{g}_{-\alpha}$, and it is nondegenerate on $\mathfrak{h}$. The roots span $\mathfrak{h}^*$, and $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}t_\alpha$. Conjugacy makes the Cartan dimension independent of the choice. Lecture 8 builds on this.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 8.1–8.3; the imports are from Sections 5.4, 6.4, 15.3 and 16.4. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapter 10. Exercise Sheet 7 gives further calculations and proofs.